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Activity 01

Q.1) Let $A = \rho \sin \phi \mathbf{a}_\rho + \rho \cos \phi \mathbf{a}_\phi$. Prove Stokes Theorem

$$\int (\nabla \times A) \cdot d\mathbf{S} = \oint_L A \cdot d\mathbf{l}.$$

Ans: $A(\rho, \phi) = \rho \sin \phi \mathbf{a}_\rho + \rho \cos \phi \mathbf{a}_\phi$

$$1 \leq \rho \leq 2, \quad 0 \leq \phi \leq \pi/2$$

computing $\nabla \times A$, $(\nabla \times A) = \frac{1}{\rho} \begin{vmatrix} \mathbf{a}_\rho & \rho \mathbf{a}_\phi & \mathbf{a}_z \\ \partial/\partial \rho & \partial/\partial \phi & \partial/\partial z \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$

$$= \frac{1}{\rho} \left(\mathbf{a}_\rho (0 - 0) - \rho \mathbf{a}_\phi (0 - 0) + \mathbf{a}_z (2\rho \cos \phi - \rho \cos \phi) \right)$$
$$= \frac{1}{\rho} (\mathbf{a}_z (\rho \cos \phi))$$

$$\nabla \times A = \cos \phi \mathbf{a}_z$$

Surface Integral: $\iint_S \cos \phi \cdot d\mathbf{S}$

$$d\mathbf{S} = -\rho d\rho d\phi \mathbf{a}_z$$

$$= \int_0^{\pi/2} \int_1^2 -\rho \cos \phi d\phi d\rho = - \left(\frac{\rho^2}{2} \right) \Big|_1^2$$

$$= - \left(\frac{4-1}{2} \right) = -\frac{3}{2}$$

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$$= \cos \phi \int_0^{\pi/2} \left. \frac{r^2}{2} \right|_1^2 d\phi \Rightarrow \cos \phi \int_0^{\pi/2} \frac{4}{2} - \frac{1}{2} d\phi \Rightarrow \cos \phi \int_0^{\pi/2} \frac{3}{2} d\phi$$

$$\Rightarrow \frac{3}{2} \int_0^{\pi/2} \cos \phi \cdot d\phi \Rightarrow \frac{3}{2} \left| \sin \phi \right|_0^{\pi/2} = \frac{3}{2}$$

Linear Integral: Outer Arc ($\rho = 2$, $\theta : \frac{\pi}{2} \rightarrow 0$)

$$A = 2 \sin \phi a_\rho + 2 \cos \phi a_\phi, \quad dl = 2 d\phi a_\phi$$

$$A \cdot dl = 4 \cos \phi d\phi \Rightarrow 4 \int_{\pi/2}^0 \cos \phi \cdot d\phi = 4 \Rightarrow \left| 4 \sin \phi \right|_{\pi/2}^0 = \boxed{-4}$$

$$dl = d\rho a_\rho, \quad A = \rho a_\rho$$

$$A \cdot dl = \rho d\rho \Rightarrow \int_1^2 \rho a_\rho = \left. \frac{\rho^2}{2} \right|_1^2 \Rightarrow \frac{3}{2}$$

Inner Arc ($\rho = 1$, $\theta : 0 \rightarrow \pi/2$)

$$dl = d\phi a_\phi, \quad \bar{A} = \sin \phi a_\rho + \cos \phi a_\phi$$

$$\bar{A} \cdot \bar{dl} = \cos \phi d\phi \Rightarrow \int_0^{\pi/2} \cos \phi d\phi \Rightarrow \left| \sin \phi \right|_0^{\pi/2} = 1$$

$$(\phi = 0, \rho = 2 \rightarrow 1) = 0$$

$$0 + 1 + \frac{3}{2} - 4 = \boxed{\frac{-3}{2}}$$

Q. 2) A vector field is specified as $A = r a_r + 3\theta a_\theta + 6\phi a_\phi$.
Find the flux of the field out of the closed surface.
defined by $0 \leq r \leq 4$, $0 \leq \theta \leq \pi/2$, $0 \leq \phi \leq \pi/2$.

$$\nabla \cdot A = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}$$

computing each term step by step.

$$\therefore \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \cdot r) = \frac{1}{r^2} \frac{\partial}{\partial r} (r^3) = \frac{3r^2}{r^2} = 3$$

$$\therefore \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (3\theta \cdot \sin \theta) = \frac{1}{r \sin \theta} 3 (\sin \theta + \theta \cos \theta) = \frac{3}{r} (1 + \theta \cot \theta)$$

$$\therefore \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi} = \frac{6}{r \sin \theta}$$

$$\nabla \cdot A = 3 + \frac{3}{r} (1 + \theta \cot \theta) + \frac{6}{r \sin \theta}$$

Volume Integral: $dV = r^2 \sin \theta dr \cdot d\theta \cdot d\phi$

$$\text{Flux} = \int_0^{\pi/2} \int_0^{\pi/2} \int_0^4 \left[3 + \frac{3}{r} (1 + \theta \cot \theta) + \frac{6}{r \sin \theta} \right] r^2 \sin \theta \cdot dr d\theta d\phi$$

$$= \int_0^{\pi/2} \int_0^{\pi/2} \int_0^4 [3r^2 \sin \theta + 3r \sin \theta (1 + \theta \cot \theta) + 6r] dr d\theta d\phi$$

$$\Rightarrow \int_0^{\pi/2} \int_0^{\pi/2} \left[r^3 \sin \theta + \frac{3}{2} r^2 (\sin \theta + \theta \cos \theta) + 3r^2 \right] \Big|_0^4 \cdot d\theta d\phi$$

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$$\int_0^{\pi/2} \int_0^{\pi/2} 88 \sin \theta + 24 \theta \cos \theta + 48 \cdot d\theta d\phi$$

$$\int_0^{\pi/2} 88 \sin \theta \cdot d\theta = 88 [-\cos \theta]_0^{\pi/2} = 88$$

$$2 - \int_0^{\pi/2} 24 \theta \cos \theta \cdot d\theta$$

$$\begin{aligned} &= \theta \int \cos \theta \cdot d\theta - \int \left(\frac{d}{d\theta} (\theta) \int \cos \theta \cdot d\theta \right) d\theta \\ &= \theta \sin \theta - \int \sin \theta \cdot d\theta \\ &= \theta \sin \theta + \cos \theta \end{aligned}$$

$$\int_0^{\pi/2} 24 [\theta \sin \theta + \cos \theta] \Big|_0^{\pi/2}$$

$$\int_0^{\pi/2} 24 [\theta \sin \theta + \cos \theta] \Big|_0^{\pi/2} = \int_0^{\pi/2} [64 + 36\pi]$$

$$[64 + 36\pi] \phi \Big|_0^{\pi/2} = [32\pi + 18\pi^2] \text{ Answer}$$